Answer Key & Teacher Notes

Session 1 — Comparing Numbers on a Number Line · Session 2 — Ordering Integers and Rational Numbers

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Comparing Numbers on a Number Line

TALK PROMPTS

- How does a point's position on the number line tell you which score is greater?
- Why is the balance with the bigger digits actually the lesser number here?

PARTNER TASK

Number Line Pairs. One partner plots −16 and −29 on a number line from −30 to 0; the other writes two true inequalities, one with < and one with >. Switch roles with the Freezer Check: −18 and 5.

WATCH FOR

- Comparing negatives by their digits (−40 > −13 because 40 > 13).
- Thinking −0.9 > −0.09 because 0.9 > 0.09.
- Confusing "farther from 0" with "greater."

SESSION 2 · TEACHER NOTES

Ordering Integers and Rational Numbers

TALK PROMPTS

- Why can one number sit farthest from zero and still be the least number in the set?
- How does sorting the negatives apart from the positives make ordering faster?

PARTNER TASK

Card Sort. One partner orders −4.5, 1.5, −0.5, and 3 from greatest to least; the other writes each absolute value. Together explain why −4.5 has the greatest absolute value but is the least number.

WATCH FOR

- Ordering negatives by their digits (writing −2, −9, −14 as least to greatest).
- Confusing greatest absolute value with greatest number.
- Ordering mixed forms without converting them first (−1/2 vs −0.75).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.C.7a	1	2	1	1	2	2	2	3	2	1	1
S1 · Form B	6.NS.C.7a	1	2	1	1	2	2	2	3	2	1	1
S2 · Form A	6.NS.C.7b	1	2	1	1	2	2	2	3	2	1	1
S2 · Form B	6.NS.C.7b	1	2	1	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

greater than (>): Farther to the **right** on the number line.

less than (<): Farther to the **left** on the number line.

inequality: A math sentence that **compares** numbers using < or >.

- On a number line, the number farther **right** is greater.
- For two negative numbers, the one closer to **0** is greater.

Example: −5 is to the **right** of −14. So −14 **<** −5.
Answer: −5 is **greater**; −14 is colder.

YOU TRY

1. a. < b. >

2. −3/4 < −1/4

EXIT TICKET

1. a. < b. <

2. Kim; −9 > −14

3. **C** (−8 < −3)

PRACTICE FORM A

1. a. > b. < c. < d. >

2. a. points at −11, −6, 0, 3 b. −6 > −11 c. greatest 3 (Tomas); least −11 (Willa)

3. −32 (−32 > −47)

4. −18 degrees (−18 < 5)

5. −24 < −9; Cave A is deeper

6. −1/8 (−7/8 < −1/8)

7. No; −0.09 is greater (−0.90 < −0.09, because −0.09 is closer to 0)

8. a. −45, −8, 12 b. −45 c. It is farthest from 0, but it is farthest left, so it is the least balance.

9. Counting order works for positive numbers. On a number line, −15 is farther left than −4, so −15 is less: −15 < −4.

10. **A** (<)

11. **B** (farther right)

Self-check word **COMPARE**

PRACTICE FORM B

1. a. > b. < c. < d. >

2. a. points at −8, −3, 0, 4 b. −3 > −8 c. greatest 4 (Eli); least −8 (Maya)

3. −28 (−28 > −53)

4. −21 degrees (−21 < 3)

5. −31 < −13; Cave B is deeper

6. −3/8 (−5/8 < −3/8)

7. No; −0.07 is greater (−0.70 < −0.07, because −0.07 is closer to 0)

8. a. −60, −9, 15 b. −60 c. It is farthest from 0, but it is farthest left, so it is the least balance.

9. Counting order works for positive numbers. On a number line, −22 is farther left than −7, so −22 is less: −22 < −7.

10. **B** (>)

11. **B** (−2)

Self-check word **NUMBERS**

GUIDED NOTES

order: To list numbers from **least** to greatest (or greatest to least).

least: The smallest number; it is farthest to the **left**.

greatest: The biggest number; it is farthest to the **right**.

- Least to greatest means **left** to right on a number line.
- Write every number in the same **form** (such as decimals) before you order.

Example: Write 3/4 as a decimal: **0.75**. Negatives first: −6 < **−0.5**. Then positives: 0.75 < 1.5. Answer: −6, −0.5, 3/4, **1.5**

YOU TRY

1. a. −8, −3, 1 b. −2, −0.2, −0.02

2. −20.25 m (the least number)

EXIT TICKET

1. a. −12, −7, 0, 3 b. −0.3, −1/4, −0.2

2. coldest −9; closest to 0: −1

3. **B** (−6.02)

PRACTICE FORM A

1. a. −14, −9, −2 b. −3, −0.5, 0, 2.5

2. a. −7, −3, 0, 2, 5 b. −7 (|−7| = 7 > 5) c. Mark (−7 is the least score)

3. −1,105 ft (order: −1,105, −640, −188)

4. −1.25, −0.75, −1/2; greatest −1/2

5. −19, −11, −2, 0, 4; coldest −19

6. 1/2, −2.05, −2.5, −3.4; greatest absolute value −3.4

7. −16.5, −16, −4 1/2, 0; deepest −16.5 ft

8. a. 1.5, −2, −0.25, 0.75, −2.5 b. −2.5, −2, −0.25, 0.75, 1.5 c. −1/4, because |−0.25| = 0.25 is the smallest absolute value

9. He ordered by distance from 0 as if the numbers were positive. For negatives, farther from 0 means farther left, so it is less. Correct order: −14, −9, −2.

10. **B** (−15)

11. **B** (−4.05)

Self-check word **GREATER**

PRACTICE FORM B

1. a. −18, −11, −5 b. −4, −0.5, 0, 1.5

2. a. −6, −2, 0, 3, 4 b. −6 (|−6| = 6 > 4) c. Nora (−6 is the least score)

3. −1,320 ft (order: −1,320, −812, −245)

4. −1.5, −3/4, −0.5; greatest −0.5

5. −15, −8, −3, 0, 6; coldest −15

6. 3/4, −1.05, −1.5, −2.6; greatest absolute value −2.6

7. −12.5, −12, −7 1/2, 0; deepest −12.5 ft

8. a. 2.5, −1, −0.75, 0.5, −1.5 b. −1.5, −1, −0.75, 0.5, 2.5 c. 0.5, because |0.5| = 0.5 is the smallest absolute value

9. He ordered by distance from 0 as if the numbers were positive. For negatives, farther from 0 means farther left, so it is less. Correct order: −12, −7, −3.

10. **B** (−13)

11. **B** (−3.06)

Self-check word **ORDERED**

SHEET 1 · COMPARING NUMBERS ON A NUMBER LINE

1. J = −4.5, K = −1, L = 3. a. J < K b. L > K c. J < L
2. a. < b. > c. < d. > e. > f. >
3. Points at −3.5 and −0.5; −3.5 < −0.5
4. a. −5 < −3 (or −3 > −5) b. Malik (−5 is the lower score)
5. Monday was colder: −7 < −2 (or −2 > −7), because −7 is farther left on the number line.
6. Yes. |−6| = 6, and −6 is left of 6 on the number line, so −6 < 6.

SHEET 2 · ORDERING INTEGERS AND RATIONAL NUMBERS

1. −3.25, −2.5, −3/4, 1.5, 2 1/4
2. a. −9, −6, 0, 4 b. −3, −0.3, −0.03
3. a. −2 b. −1.04 c. −1/2 d. 0
4. shark (−45.5), dolphin (−30.25), turtle (−12), jellyfish (−8.75)
5. He ordered by distance from 0, as if the numbers were positive, so his list goes greatest to least. For negatives, farther from 0 is less. Correct order: −0.6, −1/2, −0.4.
6. **B** (−8, −5, 0, 3)
7. **B** (−2.07)

October 2026 conceptual additions — teacher key

Session 1 · Comparing Numbers on a Number Line

−4°C is warmer and greater; −12°C is farther from zero, with absolute value 12 versus 4.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: −4°C is warmer and greater; −12°C is farther from zero, with absolute value 12 versus 4.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Ordering Integers and Rational Numbers

$-1.2 < -0.75 < -0.5 < 0$; $-3/4 = -0.75$. A more negative position lies farther left.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: $-1.2 < -0.75 < -0.5 < 0$; $-3/4 = -0.75$. A more negative position lies farther left.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.